

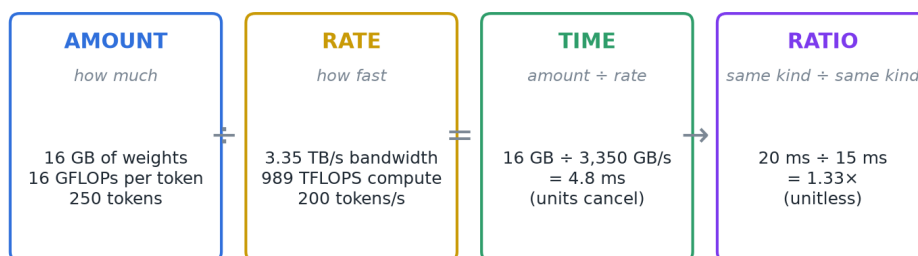
04 • Math for Memory, Throughput & Speedups

Source: [fanout.sh / inference-eng / 00-01](https://fanout.sh/inference-eng/00-01) • ~8 min

IN SHORT

Almost every number in inference is an **amount** (16 GB), a **rate** (3.35 TB/s) or a **ratio** (1.25×). With careful units and one identity, **time = amount ÷ rate**, you can answer three questions before touching hardware: does the model fit, how fast can it possibly run, and what is a claimed speedup really worth.

Three kinds of quantity, and the one identity that connects them

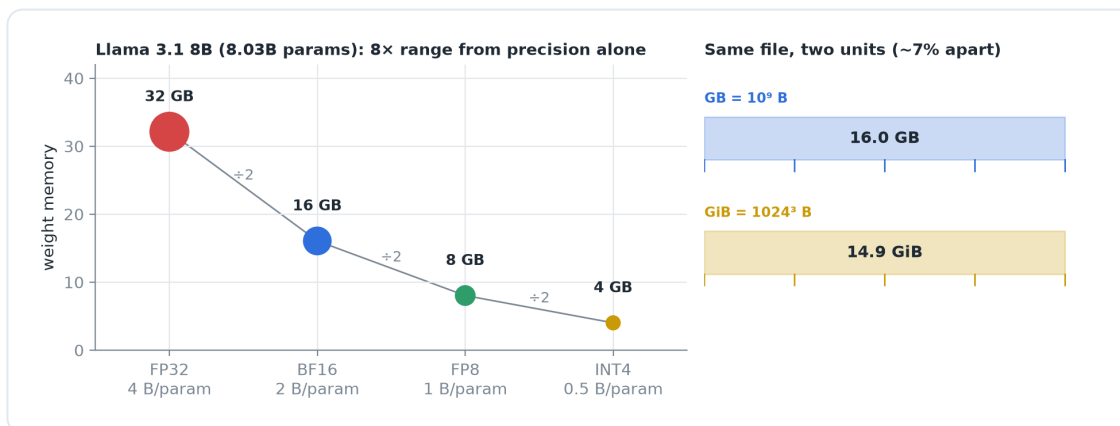


FLOPs (lowercase s) = a count, an AMOUNT: what a model costs. FLOPS (capital S) = per second, a RATE: what a GPU supplies.

Amounts divided by rates give times; times divided by times give unitless ratios.

1. Amounts: does it fit?

- Each parameter is stored at some **precision**: FP32 = **4 bytes**, BF16 = **2**, FP8 = **1**, INT4 = **0.5**.
- Weight memory = parameters × bytes per parameter**. Llama 3.1 8B has **8.03B** parameters:



Left: each halving of precision halves memory, an 8× range overall. Right: one 16 GB file reads as 14.9 GiB.

TRAP

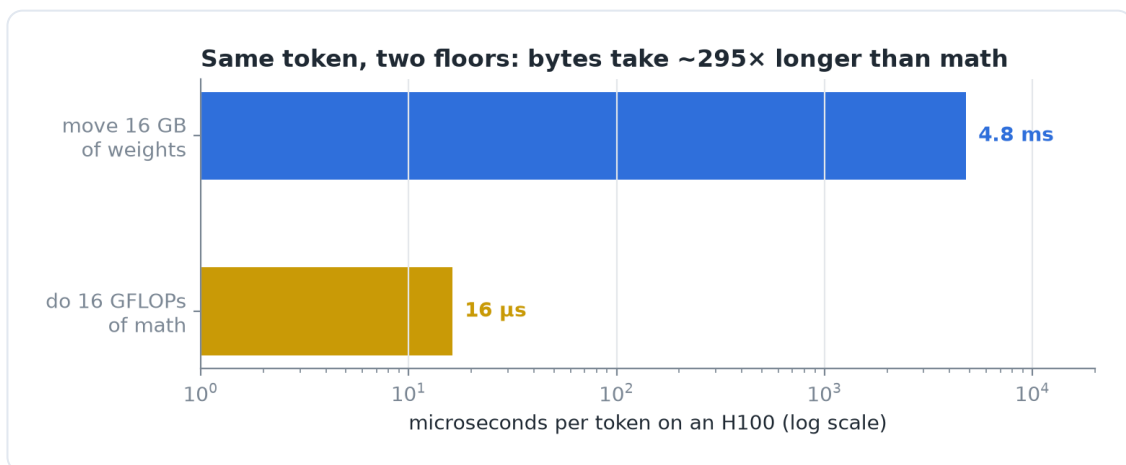
GB vs GiB: a GB is 10^9 bytes; a GiB is 1024^3 , about 7% bigger. A 16 GB file shows as **14.9 GiB**. Check the unit before doubting the math.

2. Rates: how fast can it possibly run?

Three rates do most of the work: memory bandwidth (bytes/s), compute (FLOPS) and generation speed (tokens/s).

- **FLOPs** (lowercase s) is a count: an amount a model *costs*. **FLOPS** (capital S) is per second: a rate a GPU *supplies*.
- Generating one token reads all 16 GB of weights; an H100 moves 3,350 GB/s. Put both in GB and the GB cancel:

$$t_{\text{bytes}} = \frac{16 \text{ GB}}{3350 \text{ GB/s}} \approx 4.8 \text{ ms} \quad t_{\text{math}} = \frac{2 \times 8 \times 10^9 \text{ FLOP}}{989 \times 10^{12} \text{ FLOP/s}} \approx 16 \text{ } \mu\text{s}$$

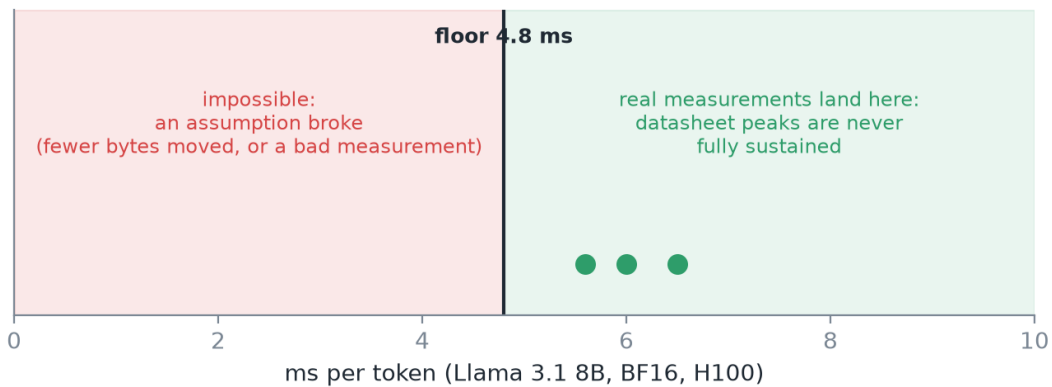


Moving the bytes takes ~300× longer than doing the math. That imbalance is what the field organizes around.

Floors, not predictions

Datasheet numbers are peaks that real systems never sustain, so these times are floors.

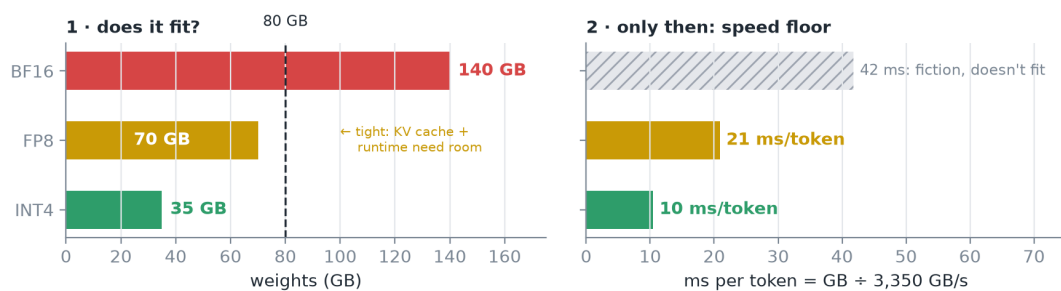
Napkin math gives a floor, not a prediction



Real measurements land above the floor. Below it means an assumption broke.

Scaling up: 70B on one 80 GB H100

70B on one H100: check the fit before the floor



BF16 doesn't fit, so its floor is meaningless. FP8 fits tightly at ~21 ms/token; INT4 fits at ~10 ms/token.

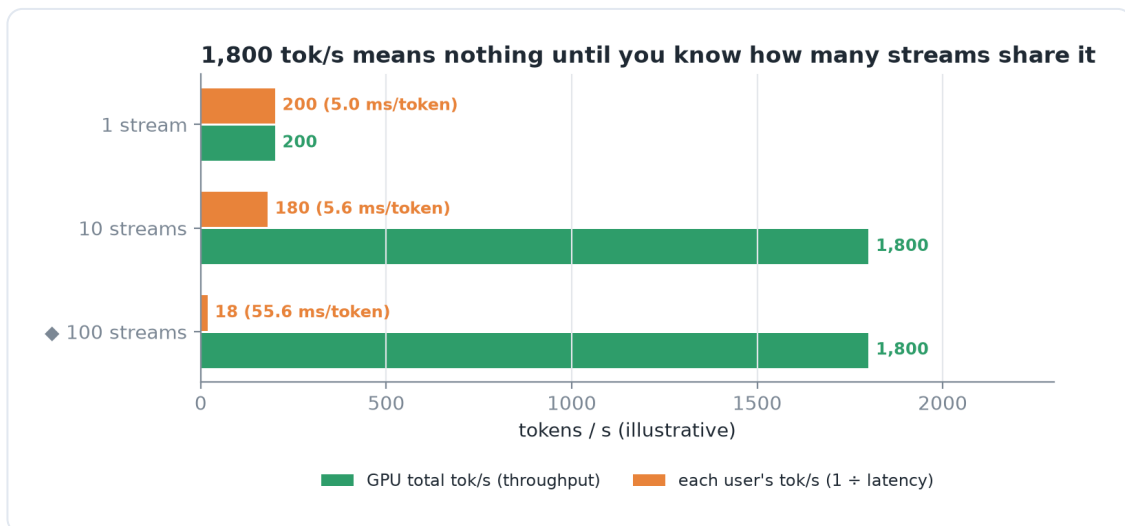
KEY IDEA

"Fits" means the **weights** fit; a running model also needs growing per-request state (KV cache) and runtime overhead, so FP8 is tight, not comfortable. And **check the fit before the floor**: a speed floor for a model that doesn't fit is fiction.

3. Latency vs throughput

	Latency	Throughput
Unit	Time per event (ms/token)	Events per second (tokens/s)
Better	Smaller	Bigger
One stream	5 ms/token	= 200 tokens/s (reciprocals)

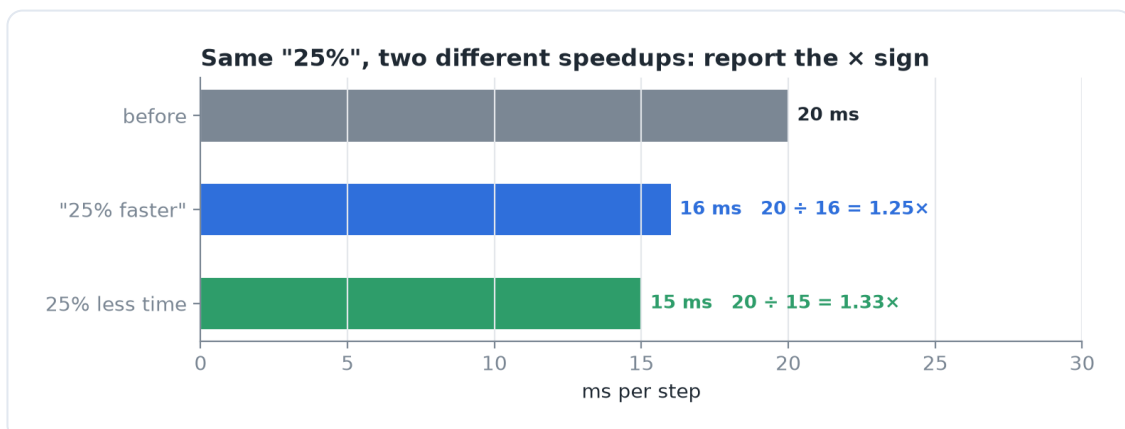
Once requests share the hardware, the two **decouple**: batching can raise total tokens/s by an order of magnitude while each stream gets no faster, and a little slower under load.



The same 1,800 tok/s can mean 180 tok/s per user or 18; the total says nothing without the stream count.

4. Speedups

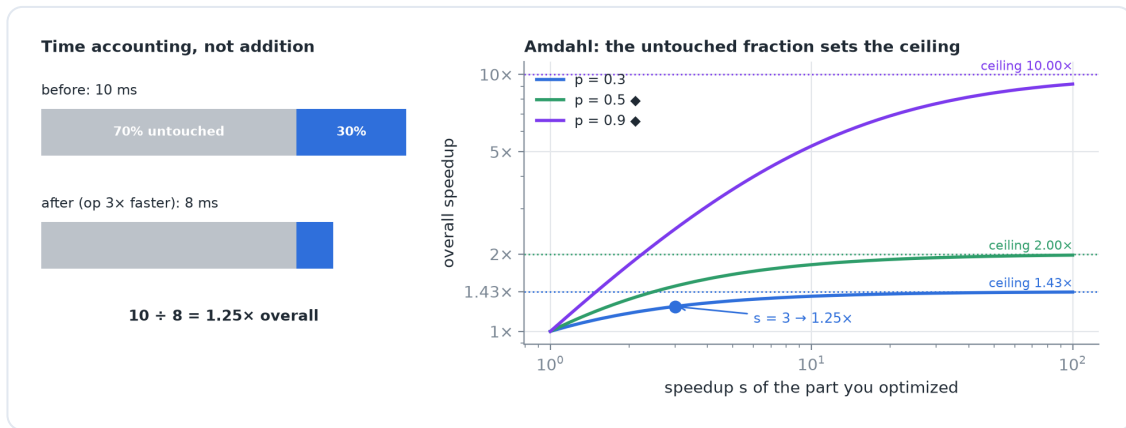
A speedup is a **ratio of times: before ÷ after**. 20 ms → 15 ms is **1.33×**.



"25% faster" usually means 1.25×, but "25% less time" is 1.33×. Report the × sign; ask which flavor a percent is.

Speedups compose by **time accounting, not addition**. If one op is 30% of a step and gets 3× faster, the 30 becomes 10 and the step drops to 80% of its old time: only **1.25×** overall. In general (**Amdahl's law**), if a fraction p of the time gets s times faster:

$$S = \frac{1}{(1 - p) + p/s} \quad \lim_{s \rightarrow \infty} S = \frac{1}{1 - p} = \frac{1}{0.7} \approx 1.43$$



Left: the 30%/70% example. Right: no matter how fast the optimized part gets, the untouched fraction caps the overall gain.

KEY IDEA

Before valuing any optimization, ask **what fraction of the time it touches**.

5. Recap

Lesson 04 on one card: amounts, rates, ratios

FIT

weights = $N \times \text{bytes/param}$
 8B: 32 / 16 / 8 / 4 GB
 check fit before floor; GB $\neq$ GiB

FLOOR

time = amount $\div$ rate
 bytes: 4.8 ms vs math: 16 μ s
 measured $\geq$ floor, always

LATENCY $\neq$ THROUGHPUT

reciprocals for one stream only
 batching raises total tok/s,
 not each user's

SPEEDUPS

before $\div$ after, report $\times$
 $1 / ((1-p) + p/s)$
 ceiling $1 / (1-p)$

Interview cheat sheet

Revise from these pages alone. ♦ = my addition beyond the video; everything else is from the lesson. Builds on 01–03.

Say it in 30 seconds

Every number is an amount, a rate or a ratio, and time = amount ÷ rate. Weight memory is params × bytes per param, so precision is the first lever for fit. Divide bytes by bandwidth and FLOPs by FLOPS to get floors; real runs land above them. Latency and throughput are reciprocals only for one stream. Speedups are before ÷ after, and Amdahl's law caps them by the fraction of time touched.

Only numbers worth memorizing

- **Bytes per param:** FP32 4, BF16 2, FP8 1, INT4 0.5.

Byte units: decimal vs binary

Decimal (SI)	Bytes	Binary (IEC)	Bytes	Binary ÷ decimal
1 KB (kilobyte)	$10^3 = 1,000$	1 KiB (kibibyte)	$2^{10} = 1,024$	1.024
1 MB (megabyte)	10^6	1 MiB (mebibyte)	$2^{20} = 1,048,576 \approx 1.049 \times 10^6$	1.049
1 GB (gigabyte)	10^9	1 GiB (gibibyte)	$2^{30} = 1,073,741,824 \approx 1.074 \times 10^9$	1.074
1 TB (terabyte)	10^{12}	1 TiB (tebibyte)	$2^{40} = 1,099,511,627,776 \approx 1.100 \times 10^{12}$	1.100

The "i" means binary: kibi = **kilo binary**, mebi = **mega binary**, and so on. Say "kibibyte" (KIB-ee-byte), "gibibyte" (GIB-ee-byte).

- **GB → GiB:** divide by 1.074. 16 GB = **14.9 GiB**; an 80 GB H100 = **74.5 GiB**.
- ♦ Datasheet rates are decimal: **1 TB/s = 1 GB/ms = 1 MB/μs**, so 3.35 TB/s moves 16 GB in $16 \div 3.35 \approx 4.8$ ms.

Derive, don't memorize

1. Weights = params × bytes per param; check fit first

$$\text{weights} = N \times \frac{\text{bits}}{8} : \quad 70\text{B} \rightarrow 140 \text{ (BF16)}, 70 \text{ (FP8)}, 35 \text{ (INT4)} \text{ GB vs } 80 \text{ GB}$$

INTUITION

"Fits" means weights + KV cache at peak + runtime (KV capacity: see 02, derivation 1). Too big: split across GPUs or quantize.

2. Time = amount ÷ rate, and the slower floor wins ♦

Each token must both move bytes and do math; whichever takes longer sets the floor:

$$t \geq \max \left(\frac{\text{bytes}}{\text{BW}}, \frac{\text{FLOPs}}{\text{FLOPS}} \right) \quad 70\text{B FP8} : \frac{70}{3350} \approx 21 \text{ ms} \quad \text{INT4} : \approx 10 \text{ ms}$$

INTUITION

A measurement below the floor means an assumption broke (fewer bytes moved, or a bad measurement). Why bytes usually win: see 01, derivation 2.

3. Speedup = before ÷ after; Amdahl caps it

$$S = \frac{t_{\text{before}}}{t_{\text{after}}} \quad S_{\text{overall}} = \frac{1}{(1-p) + p/s} \leq \frac{1}{1-p}$$

INTUITION

"x% faster" = $1 + x$; "x% less time" = $1 / (1 - x)$. 30% of a step made 3× faster gives 1.25×; infinitely faster gives 1.43×.

Symptom → diagnosis

Symptom	Diagnosis
Tool shows 14.9 for a 16 GB file	GiB vs GB, not a bug
Measured faster than the bandwidth floor	Fewer bytes moved than assumed, or a broken measurement
Kernel 3× faster, end to end barely moves	Small p : Amdahl
"1,800 tok/s" but users say it's slow	Throughput across many streams; ask per-stream speed

Rapid-fire Q&A

Question	Crisp answer
FLOPs vs FLOPS?	Count (what a model costs) vs per second (what a GPU supplies).
Why are napkin numbers floors?	Datasheet peaks are never sustained, so real runs are slower.
20 ms → 15 ms: how much faster?	1.33× (25% less time), not 1.25×.
Is a throughput gain a latency gain?	Only for one stream; with batching they decouple.

TRAP

- Adding speedups (" $2\times + 2\times = 4\times$ "): compose by time accounting instead.
- Quoting a speed floor for a model that doesn't fit.